

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (ISE) (Credit System) Degree Examinations**

May - June 2023

20ISE44 – BUSINESS INTELLIGENCE AND ITS APPLICATIONS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define business intelligence in information technology. Explain the Malcolm Baldrige performance excellence framework.	10	L*2	1	1, 2
	b) Explain the typical enterprise application architecture with example and one case study.	10	L2	1	1, 2
2.	a) Explain the various methods of storing unstructured data with challenges and possible solutions.	10	L2	2	1, 2
	b) Explain the methods of managing semi-structured data.	4	L2	2	1, 2
	c) Explain how XML is a possible solution for semi-structured data management with an example.	6	L2	2	1, 2
3.	a) Explain OLAP in business intelligence with a diagram.	10	L2	2	1, 2
	b) Explain few OLAP operations on multidimensional data with an example for each.	10	L2	2	2
Unit – II					
4.	a) Define decision making. Explain strategic, tactical and operational decisions by providing appropriate examples.	10	L2	3	1, 2
	b) Explain how retail industry leverages the business intelligence with an appropriate example.	10	L2	3	1, 2
5.	a) Explain the business intelligence framework with an architecture diagram.	10	L2	3	1, 2
	b) Explain the different types of business intelligence users with an example.	5	L2	3	1, 2
	c) Explain the data integration process and its challenges.	5	L2	3	1, 2
6.	a) Define ETL. Explain the various stages in ETL process with an architecture diagram.	10	L2	4	1, 2
	b) Explain the following in ETL process: i. Data Mapping ii. Data Staging iii. Data Extraction	6	L2	4	1, 2
	c) Explain the star schema with a figure.	4	L2	4	1, 2
Unit – III					
7.	a) List and explain few attributes of a good metric.	5	L2	5	1, 2
	b) Explain the enterprise reporting characteristics in OLAP world.	5	L2	5	1, 2
	c) Explain the steps and benefits of creating an enterprise dashboard.	10	L2	5	1, 2
8.	a) Explain the evolution of traditional business intelligence.	10	L2	5	1, 2
	b) Explain the cloud based business intelligence technology in detail.	10	L2	5	1, 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
